Supplementary Materials

A Proofs

A.1 Proof of Theorem 1

Proof. By their construction in Definition 1, the random variables $(X^{(1)}, \ldots, X^{(K)}) \sim Q_\theta^{(1)} \times \cdots \times Q_\theta^{(K)}$ have conditional distribution

$$(X^{(1)}, \ldots, X^{(K)}) | \{X = t\} \sim G_t.$$

Furthermore, Definition 1 tells us that $X = T(X^{(1)}, \ldots, X^{(K)})$. This means that

$$(X^{(1)}, \ldots, X^{(K)}) | \{T(X^{(1)}, \ldots, X^{(K)}) = t\} \sim G_t,$$

which establishes part (b) of the theorem.

The distribution G_t in Definition 1 does not depend on θ (note that it is associated with the entire family $\mathcal{P}$, not a particular distribution P_θ). By the definition of sufficiency, the fact that the conditional distribution $(X^{(1)}, \ldots, X^{(K)}) | T(X^{(1)}, \ldots, X^{(K)})$ does not depend on θ implies that $T(X^{(1)}, \ldots, X^{(K)})$ is sufficient for θ. This proves (a). □

A.2 Proof of Theorem 2

Proof. We start by proving the $\Longleftarrow$ direction. Suppose H is the convolution of $H_1, \ldots, H_K$. We follow the recipe given in Algorithm 1:

1. We choose $\mathcal{Q}^{(k)} = \mathcal{P}^{H_k}$ for $k = 1, \ldots, K$.

2. Let $(X^{(1)}, \ldots, X^{(K)}) \sim P_\theta^{H_1} \times \cdots \times P_\theta^{H_K}$. This joint distribution satisfies

$$\prod_{k=1}^{K} dP_\theta^{H_k}(x^{(k)}) = \exp\left\{ \left(\sum_{k=1}^{K} x^{(k)} \right)^\top \theta - \sum_{k=1}^{K} \psi_{H_k}(\theta) \right\} \prod_{k=1}^{K} dH_k(x^{(k)}).$$

   By the factorization theorem, we find that $T(X^{(1)}, \ldots, X^{(K)}) = \sum_{k=1}^{K} X^{(k)}$ is sufficient for θ.

3. It remains to determine the distribution of $U = T(X^{(1)}, \ldots, X^{(K)})$. This random variable is the convolution of $P_\theta^{H_1} \times \cdots \times P_\theta^{H_K}$, and its distribution μ is defined by the

following K-way integral:

$$\begin{aligned} d\mu(u) &= \int \cdots \int 1\left\{\sum_{k=1}^{K} x^{(k)} = u\right\} \prod_{k=1}^{K} dP_\theta^{H_k}(x^{(k)}) \\ &= \exp\left\{u^\top \theta - \sum_{k=1}^{K} \psi_{H_k}(\theta)\right\} \int \cdots \int 1\left\{\sum_{k=1}^{K} x^{(k)} = u\right\} \prod_{k=1}^{K} dH_k(x^{(k)}) \\ &= \exp\left\{u^\top \theta - \psi_H(\theta)\right\} dH(u) \\ &= dP_\theta^H(u), \end{aligned}$$

where in the second-to-last equality we use the assumption that H is the K-way convolution of $H_1, \ldots, H_K$ and the fact that the moment generating function of a convolution is the product of the individual moment generating functions (and recalling that ψ_H is the logarithm of the moment generating function of H). This establishes that $T(X^{(1)}, \ldots, X^{(K)}) \sim P_\theta^H$. By Theorem 1, the family $\mathcal{P}^H$ is thinned by this choice of $T(\cdot)$.

We now prove the $\implies$ direction. Suppose that $\mathcal{P}^H$ can be K-way thinned into $\mathcal{P}^{H_1}, \ldots, \mathcal{P}^{H_K}$ using the summation function. Then applying Definition 1 with $\theta = 0$, we can take $X \sim P_0^H$ and produce $(X^{(1)}, \ldots, X^{(K)}) \sim P_0^{H_1} \times \cdots \times P_0^{H_K}$ for which $X = X^{(1)} + \cdots + X^{(K)}$. Noting that $P_0^{H_k} = H_k$ for all k and $P_0^H = H$, this proves that H is a K-way convolution of $H_1, \ldots, H_K$. □

A.3 Proof of Theorem 3

Proof. Suppose that $X \sim P_\theta$ is a natural exponential family with d-dimensional parameter θ that can be thinned by $T(\cdot)$ into $X^{(k)} \overset{ind}{\sim} Q_\theta^{(k)}$ for $k = 1, \ldots K$. By Theorem 1, $T(X^{(1)}, \ldots, X^{(K)})$ is a sufficient statistic for θ on the basis of $X^{(1)}, \ldots, X^{(K)}$, which implies that the conditional distribution $(X^{(1)}, \ldots, X^{(K)}) | T(X^{(1)}, \ldots, X^{(K)}) = t$ does not depend on θ. We can write the conditional density with respect to the appropriate dominating measure as

$$\begin{aligned} &f_{X^{(1)},\ldots,X^{(K)}|T(X^{(1)},\ldots,X^{(K)})=t}(x^{(1)}, \ldots, x^{(K)}) \\ &= \frac{f_{X^{(1)},\ldots,X^{(K)}}(x^{(1)}, \ldots, x^{(K)}) 1\{T(x^{(1)}, \ldots, x^{(K)}) = t\}}{f_{T(X^{(1)},\ldots,X^{(K)})}(t)} \\ &= \frac{q_\theta^{(1)}(x^{(1)}) \ldots q_\theta^{(K)}(x^{(K)}) 1\{T(x^{(1)}, \ldots, x^{(K)}) = t\}}{\exp(T(x^{(1)}, \ldots, x^{(K)})^\top \theta - \psi(\theta)) h(T(x^{(1)}, \ldots, x^{(K)}))} \\ &= \frac{\prod_{k=1}^{K} q_\theta^{(k)}(x^{(k)})}{\exp(T(x^{(1)}, \ldots, x^{(K)})^\top \theta - \psi(\theta))} \cdot \frac{1\{T(x^{(1)}, \ldots, x^{(K)}) = t\}}{h(T(x^{(1)}, \ldots, x^{(K)}))}, \end{aligned}$$

where in the second equality, we used that $T(X^{(1)}, \ldots, X^{(K)}) \overset{D}{=} X \sim P_\theta$.

As this distribution cannot depend on θ, the first fraction must be constant in θ. That is, for any fixed $\theta_0 \in \Omega$,

$$\begin{aligned}
&\frac{\prod_{k=1}^K q_\theta^{(k)}(x^{(k)})}{\exp(T(x^{(1)},\dots,x^{(K)})^\top\theta - \psi(\theta))} = \frac{\prod_{k=1}^K q_{\theta_0}^{(k)}(x^{(k)})}{\exp(T(x^{(1)},\dots,x^{(K)})^\top\theta_0 - \psi(\theta_0))} \\
\iff &\prod_{k=1}^K \frac{q_\theta^{(k)}(x^{(k)})}{q_{\theta_0}^{(k)}(x^{(k)})} = \exp(T(x^{(1)},\dots,x^{(K)})^\top(\theta-\theta_0) - (\psi(\theta)-\psi(\theta_0))) \\
\iff &T(x^{(1)},\dots,x^{(K)})^\top(\theta-\theta_0) = \sum_{k=1}^K \left[\log q_\theta^{(k)}(x^{(k)}) + \frac{1}{K}\psi(\theta) - \log q_{\theta_0}^{(k)}(x^{(k)}) - \frac{1}{K}\psi(\theta_0)\right].
\end{aligned} \tag{5}$$

To proceed, we must first confirm that the term inside the summation on the right-hand side is linear in $\theta - \theta_0$. To see this, observe that if we replace $x^{(1)}$ with $\tilde{x}^{(1)}$, then

$$\begin{aligned}
&\left[T(x^{(1)},x^{(2)},\dots,x^{(K)}) - T(\tilde{x}^{(1)},x^{(2)},\dots,x^{(K)})\right]^\top(\theta-\theta_0) \\
&= \log q_\theta^{(1)}(x^{(1)}) - \log q_{\theta_0}^{(1)}(x^{(1)}) - \log q_\theta^{(1)}(\tilde{x}^{(1)}) + \log q_{\theta_0}^{(1)}(\tilde{x}^{(1)}) \\
&= a^{(1)}(x^{(1)},\theta) - a^{(1)}(x^{(1)},\theta_0) - a^{(1)}(\tilde{x}^{(1)},\theta) + a^{(1)}(\tilde{x}^{(1)},\theta_0)
\end{aligned}$$

where $a^{(1)}(x,\theta) = \log q_\theta^{(1)}(x)$. Since the initial expression in the previous string of equalities is linear in θ, the same must be true for the final expression, implying that $a^{(1)}$ must be of the form

$$a^{(1)}(x,\theta) = T^{(1)}(x)^\top\theta + f^{(1)}(x) + g^{(1)}(\theta)$$

for some functions $T^{(1)}(\cdot)$, $f^{(1)}(\cdot)$, and $g^{(1)}(\cdot)$.

Substituting into the above,

$$\left[T(x^{(1)},x^{(2)},\dots,x^{(K)}) - T(\tilde{x}^{(1)},x^{(2)},\dots,x^{(K)})\right]^\top(\theta-\theta_0) = \left[T^{(1)}(x^{(1)}) - T^{(1)}(\tilde{x}^{(1)})\right]^\top(\theta-\theta_0).$$

Applying the same logic to every $k = 1,\dots,K$ in sequence, we have that for any k,

$$\begin{aligned}
&\left[T(\tilde{x}^{(1)},\dots,\tilde{x}^{(k-1)},x^{(k)},x^{(k+1)},\dots,x^{(K)}) - T(\tilde{x}^{(1)},\dots,\tilde{x}^{(k-1)},\tilde{x}^{(k)},x^{(k+1)},\dots,x^{(K)})\right]^\top(\theta-\theta_0) \\
&= \left[T^{(k)}(x^{(k)}) - T^{(k)}(\tilde{x}^{(k)})\right]^\top(\theta-\theta_0)
\end{aligned}$$

for some function $T^{(k)}(\cdot)$.

Summing over $k = 1,\dots,K$ then yields

$$\left[T(x^{(1)},\dots,x^{(K)}) - T(\tilde{x}^{(1)},\dots,\tilde{x}^{(K)})\right]^\top(\theta-\theta_0) = \left[\sum_{k=1}^K T^{(k)}(x^{(k)}) - \sum_{k=1}^K T^{(k)}(\tilde{x}^{(k)})\right]^\top(\theta-\theta_0).$$

Since $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ is a d-dimensional full-rank natural exponential family, there exists a $\theta_0 \in \Omega$ and $\epsilon > 0$ such that $\theta = \theta_0 + \epsilon v \in \Omega$ for every $v \in \mathbb{R}^d$ such that $\|v\|_2 = 1$.

Since the previous display is true for every pair of θ and θ_0, selecting pairs such that $\theta - \theta_0 = \epsilon v$ simplifies the above into

$$\left[T(x^{(1)}, \ldots, x^{(K)}) - T(\tilde{x}^{(1)}, \ldots, \tilde{x}^{(K)})\right]^\top v = \left[\sum_{k=1}^{K} T^{(k)}(x^{(k)}) - \sum_{k=1}^{K} T^{(k)}(\tilde{x}^{(k)})\right]^\top v.$$

As the above holds for all $v \in \mathbb{R}^d$ such that $||v||_2 = 1$, restricting our attention to the standard basis vectors implies that

$$T(x^{(1)}, \ldots, x^{(K)}) - T(\tilde{x}^{(1)}, \ldots, \tilde{x}^{(K)}) = \sum_{k=1}^{K} T^{(k)}(x^{(k)}) - \sum_{k=1}^{K} T^{(k)}(\tilde{x}^{(k)})$$

and furthermore that

$$T(x^{(1)}, \ldots, x^{(K)}) = \sum_{k=1}^{K} T^{(k)}(x^{(k)}) + c.$$

Without loss of generality, c can be absorbed into the $T^{(k)}(\cdot)$ functions, thus proving the claim that if a natural exponential family can be thinned, then the function $T(\cdot)$ must be a summation of the form $T(X^{(1)}, \ldots, X^{(K)}) = \sum_{k=1}^{K} T^{(k)}(X^{(k)})$ for some functions $T^{(k)}(\cdot)$ for $k = 1, \ldots, K$.

Finally, plugging this expression into (5) gives

$$\sum_{k=1}^{K} T^{(k)}(x^{(k)})^\top(\theta - \theta_0) = \sum_{k=1}^{K} \left[\log q_\theta^{(k)}(x^{(k)}) + \frac{1}{K}\psi(\theta) - \log q_{\theta_0}^{(k)}(x^{(k)}) - \frac{1}{K}\psi(\theta_0)\right],$$

which shows that the functions $q_\theta^{(k)}(\cdot)$ can be characterised as

$$\begin{aligned}
& T^{(k)}(x^{(k)})^\top(\theta - \theta_0) = \log q_\theta^{(k)}(x^{(k)}) + \frac{1}{K}\psi(\theta) - \log q_{\theta_0}^{(k)}(x^{(k)}) - \frac{1}{K}\psi(\theta_0) \\
\iff & \log q_\theta^{(k)}(x^{(k)}) = T^{(k)}(x^{(k)})^\top(\theta - \theta_0) - \frac{1}{K}\psi(\theta) + \log q_{\theta_0}^{(k)}(x^{(k)}) + \frac{1}{K}\psi(\theta_0) \\
\iff & q_\theta^{(k)}(x^{(k)}) = q_{\theta_0}^{(k)}(x^{(k)}) \exp\left(T^{(k)}(x^{(k)})^\top(\theta - \theta_0) - \frac{1}{K}\psi(\theta) + \frac{1}{K}\psi(\theta_0)\right).
\end{aligned}$$

Thus, $q_\theta^{(k)}(\cdot)$ is the density of an exponential family with sufficient statistic $T^{(k)}(\cdot)$ and carrier density given by $h^{(k)}(x^{(k)}) \propto q_{\theta_0}^{(k)}(x^{(k)}) \exp(-T^{(k)}(x^{(k)})^\top\theta_0)$.

□

A.4 Proof of Proposition 2

Proof. The result follows from a chain of equalities:

$$\begin{aligned} I_X(\theta) &= I_{S(X)}(\theta) \\ &= I_{T(X^{(1)},\ldots,X^{(K)})}(\theta) \\ &= I_{(X^{(1)},\ldots,X^{(K)})}(\theta) \\ &= \sum_{k=1}^{K} I_{X^{(k)}}(\theta). \end{aligned}$$

The first equality is true because $S(X)$ is sufficient for θ based on X. The second equality follows from the definition of thinning $S(X)$ into $X^{(1)}, \ldots, X^{(K)}$ using $T(\cdot)$. The third equality follows from Theorem 1, which tells us that $T(X^{(1)}, \ldots, X^{(K)})$ is sufficient for θ based on $(X^{(1)}, \ldots, X^{(K)})$. The final equality follows from independence. □

A.5 Proof of Theorem 4

Proof. We begin by providing some intuition. Since $Z^{(1)}$ and $Z^{(2)}$ are non-constant random variables, their supports each must contain more than one element. Therefore, by independence, the support of $Z^{(1)} + Z^{(2)}$ must contain more than two elements and thus cannot be Bernoulli.

Formally, let $Q^{(1)}$ and $Q^{(2)}$ be the distributions of $Z^{(1)}$ and $Z^{(2)}$, respectively. If $Z^{(1)}+Z^{(2)}$ were Bernoulli, then

$$\begin{aligned} 1 &= \mathbb{P}(Z^{(1)} + Z^{(2)} \in \{0, 1\}) \\ &= \int \mathbb{P}(Z^{(2)} \in \{0 - z^{(1)},\ 1 - z^{(1)}\} | Z^{(1)} = z^{(1)}) dQ^{(1)}(z^{(1)}) \\ &= \int \mathbb{P}(Z^{(2)} \in \{0 - z^{(1)},\ 1 - z^{(1)}\}) dQ^{(1)}(z^{(1)}), \end{aligned}$$

where the last equality follows by independence of $Z^{(1)}$ and $Z^{(2)}$. For this integral to equal 1, we would need

$$\mathbb{P}(Z^{(2)} \in \{0 - z^{(1)},\ 1 - z^{(1)}\}) = 1 \text{ for } Q^{(1)}\text{-almost every } z^{(1)} \tag{6}$$

since $P(Z^{(2)} \in \{0 - z^{(1)},\ 1 - z^{(1)}\})$ is bounded above by 1. For $Z^{(1)}$ to be non-constant, there must be at least two distinct points a and b such that (6) holds with $z^{(1)} = a$ and holds with $z^{(1)} = b$. Since the intersection of two probability 1 sets is a set that holds with probability 1, we have that

$$\mathbb{P}(Z^{(2)} \in \{-a,\ 1 - a\} \cap \{-b,\ 1 - b\}) = 1,$$

from which it follows that $\{-a,\ 1 - a\} \cap \{-b,\ 1 - b\}$ is non-empty. However, there is no choice of $a \neq b$ for which this intersection has more than one element (which is required for $Z^{(2)}$ to be non-constant). Thus we arrive at a contradiction.

□

A.6 Proof of Corollary 2

Proof. Since the Bernoulli family is a natural exponential family, if at least one of the conclusions of Theorem 3 is always false for the Bernoulli, then the contrapositive of Theorem 3 will imply that the Bernoulli distribution cannot be thinned by any function $T(\cdot)$.

Suppose that $X \sim \text{Bernoulli}(\theta)$. Consider the first conclusion, namely that the thinning function $T(x^{(1)}, \ldots, x^{(K)})$ is of the form $\sum_{k=1}^{K} T^{(k)}(x^{(k)})$. This would imply that $\sum_{k=1}^{K} T^{(k)}(X^{(k)}) = X \sim \text{Bernoulli}(\theta)$. However, by Theorem 4, $T(\cdot)$ cannot be a convolution of independent, non-constant random variables. Therefore, the second conclusion can only be true if $X^{(1)}, \ldots, X^{(K)}$ are not mutually independent, some or all of $X^{(1)}, \ldots, X^{(K)}$ are constant, or some or all of $T^{(1)}(\cdot), \ldots, T^{(K)}(\cdot)$ are constant functions. All three cases violate the second conclusion of Theorem 3 that $X^{(k)}$ are independent exponential families. Therefore, both conclusions of Theorem 3 cannot be simultaneously true, thus proving the claim. □

B Connecting Example 3.1 to prior work

Other authors have considered obtaining two independent Gaussian random variables $\mathbf{U}$ and $\mathbf{V}$ from a single Gaussian random variable $\mathbf{X} \sim N_n(\boldsymbol{\theta}, \mathbf{I}_n)$ by generating $\mathbf{W} \sim N_n(\mathbf{0}_n, \gamma \mathbf{I}_n)$ for a tuning parameter $\gamma > 0$, and then setting $\mathbf{U} = \mathbf{X} + \mathbf{W}$ and $\mathbf{V} = \mathbf{X} - \gamma^{-1}\mathbf{W}$. Then, $\mathbf{U} \sim N_n(\boldsymbol{\theta}, (1+\gamma)\mathbf{I}_n)$ and $\mathbf{V} \sim N_n(\boldsymbol{\theta}, (1+\gamma^{-1})\mathbf{I}_n)$ are independent. Rasines and Young (2022) and Leiner *et al.* (2023) applied this decomposition to address Scenario 1 in Section 1. Additionally, Rasines and Young (2022) showed that this leads to asymptotically valid inference under certain regularity conditions, even when $\mathbf{X}$ is not normally distributed. Tian (2020) and Oliveira *et al.* (2021) applied this decomposition to address Scenario 2 in Section 1.

This decomposition is in fact identical to Example 3.1 up to scaling, with $\mathbf{X}^{(1)} = \epsilon_1 \mathbf{U}$, $\mathbf{X}^{(2)} = \epsilon_2 \mathbf{V}$, $\epsilon_1 = (1+\gamma)^{-1}$, and $\epsilon_2 = 1 - \epsilon_1$. In particular, to thin $\mathbf{X} \sim N_n(\boldsymbol{\theta}, \mathbf{I}_n)$ by addition into $(\mathbf{X}^{(1)}, \mathbf{X}^{(2)})$, we sample from $G_{\mathbf{X}}$ where $G_{\mathbf{t}}$, defined in Theorem 1, can be shown to equal the singular multivariate normal distribution

$$N_{2n}\left(\begin{pmatrix}\epsilon_1 \mathbf{t} \\ \epsilon_2 \mathbf{t}\end{pmatrix}, \epsilon_1\epsilon_2 \begin{pmatrix}\mathbf{I}_n & -\mathbf{I}_n \\ -\mathbf{I}_n & \mathbf{I}_n\end{pmatrix}\right).$$

Sampling from $G_{\mathbf{X}}$ is equivalent to sampling $\mathbf{W} \sim N_n(\mathbf{0}_n, \gamma \mathbf{I}_n)$ (independent of $\mathbf{X}$) and then generating $\mathbf{X}^{(1)} = \epsilon_1(\mathbf{X} + \mathbf{W})$ and $\mathbf{X}^{(2)} = \epsilon_2(\mathbf{X} - \gamma^{-1}\mathbf{W})$. These ideas can easily be generalized to thin $\mathbf{X} \sim N_n(\boldsymbol{\theta}, \boldsymbol{\Sigma})$ with a known positive definite covariance matrix $\boldsymbol{\Sigma}$.

C Derivations of thinning procedures

C.1 Exponential families

C.1.1 Weibull distribution

The gamma family with known shape α and unknown rate θ admits a collection of thinning functions, indexed by a hyperparameter $\nu > 0$, that thin the gamma family into the Weibull family.

Example C.1 (Thinning Gamma(α, θ) with $\alpha = K$ known, approach 3)**.** *Recall that the Weibull distribution with known shape parameter ν (but varying scale λ) is a general exponential family. Then, starting with $X^{(k)} \overset{iid}{\sim}$ Weibull(λ, ν) for $k = 1, \ldots, K$, we have that $T^{(k)}(x^{(k)}) = (x^{(k)})^\nu$. We can thus apply Proposition 1 using the function*

$$T(x^{(1)}, \ldots, x^{(K)}) = \sum_{k=1}^{K} (x^{(k)})^\nu$$

to thin the distribution of $\sum_{k=1}^{K} (X^{(k)})^\nu$ into $(X^{(1)}, \ldots, X^{(K)})$. As $\sum_{k=1}^{K} (X^{(k)})^\nu \sim$ Gamma$(K, \lambda^{-\nu})$, taking $\lambda = \theta^{-1/\nu}$ yields the desired result.

To generate $(X^{(1)}, \ldots, X^{(K)})$, we can first apply the K-fold gamma thinning result discussed in Example 3.2 with $\epsilon_k = \frac{1}{K}$ to generate $Y^{(k)} \overset{iid}{\sim}$ Exp$(\lambda^{-\nu})$, and then compute $X^{(k)} = (Y^{(k)})^{\frac{1}{\nu}}$.

Proof of Example C.1. We must prove that if $X^{(k)} \overset{iid}{\sim}$ Weibull(λ, ν), for $k = 1, \ldots, K$, then $\sum_{k=1}^{K} \left(X^{(k)}\right)^\nu \sim$ Gamma$(K, \lambda^{-\nu})$.

Recalling that the gamma distribution is convolution-closed in its shape parameter, it is sufficient to show for a single $X^{(k)} \sim$ Weibull(λ, ν) random variable that $(X^{(k)})^\nu \sim$ Gamma$(1, \lambda^{-\nu}) =$ Exp$(\lambda^{-\nu})$, where $\nu > 0$. Denote $Z = (X^{(k)})^\nu$. Then,

$$\begin{aligned} f_Z(z) &= f_{X^{(k)}}\left(z^{\frac{1}{\nu}}\right) \left|\frac{dx^{(k)}}{dz}\right| \\ &\propto \left(z^{\frac{1}{\nu}}\right)^{\nu-1} \exp\left(-\left(\frac{z^{\frac{1}{\nu}}}{\lambda}\right)^\nu\right) \left|\frac{1}{\nu} z^{-\frac{\nu-1}{\nu}}\right| \\ &\propto \exp\left(-\lambda^{-\nu} z\right). \end{aligned}$$

The above implies that $(X^{(k)})^\nu \sim$ Exp$(\lambda^{-\nu})$, and thus $\sum_{k=1}^{K} \left(X^{(k)}\right)^\nu \sim$ Gamma$(K, \lambda^{-\nu})$ as required.

□

C.1.2 Beta distribution

Proof of Example 4.1. We must prove the following three claims:

1. *If* $X^{(k)} \sim Beta\left(\frac{1}{K}\theta + \frac{k-1}{K}, \frac{1}{K}\beta\right)$*, for* $k = 1, \ldots, K$*, and* $X^{(k)}$ *are mutually independent, then* $\left(\prod_{k=1}^{K} X^{(k)}\right)^{\frac{1}{K}} \sim Beta(\theta, \beta)$*.*

   Recall that the beta distribution is fully characterised by its moments due to its finite support (Feller, 1971). Also recall that the expectation of the rth power of a $X \sim \text{Beta}(\theta, \beta)$ random variable is $E[X^r] = \frac{B(\theta+r,\beta)}{B(\theta,\beta)}$ where B is the beta function. Finally, note that the Gauss multiplication theorem (Abramowitz *et al.*, 1972, page 256) says that

$$\prod_{k=1}^{K} \Gamma\left(z + \frac{k-1}{K}\right) = (2\pi)^{\frac{K-1}{2}} K^{\frac{1}{2}-Kz} \Gamma(Kz).$$

   Then, the rth moment of $\left(\prod_{k=1}^{K} X^{(k)}\right)^{\frac{1}{K}}$ is

$$\begin{aligned}
E\left[\left(\left(\prod_{k=1}^{K} X^{(k)}\right)^{\frac{1}{K}}\right)^{r}\right] &= \prod_{k=1}^{K} E\left[\left(X^{(k)}\right)^{\frac{r}{K}}\right] \\
&= \prod_{k=1}^{K} \frac{B(\frac{1}{K}\theta + \frac{k-1}{K} + \frac{r}{K}, \frac{1}{K}\beta)}{B(\frac{1}{K}\theta + \frac{k-1}{K}, \frac{1}{K}\beta)} \\
&= \prod_{k=1}^{K} \frac{\frac{\Gamma(\frac{1}{K}\theta + \frac{k-1}{K} + \frac{r}{K})\Gamma(\frac{1}{K}\beta)}{\Gamma(\frac{1}{K}\theta + \frac{k-1}{K} + \frac{r}{K} + \frac{1}{K}\beta)}}{\frac{\Gamma(\frac{1}{K}\theta + \frac{k-1}{K})\Gamma(\frac{1}{K}\beta)}{\Gamma(\frac{1}{K}\theta + \frac{k-1}{K} + \frac{1}{K}\beta)}} \\
&= \prod_{k=0}^{K-1} \frac{\Gamma(\frac{1}{K}\theta + \frac{r}{K} + \frac{k}{K})\Gamma(\frac{1}{K}\theta + \frac{1}{K}\beta + \frac{k}{K})}{\Gamma(\frac{1}{K}\theta + \frac{1}{K}\beta + \frac{r}{K} + \frac{k}{K})\Gamma(\frac{1}{K}\theta + \frac{k}{K})} \\
&= \frac{\left[\prod_{k=0}^{K-1} \Gamma(\frac{1}{K}\theta + \frac{r}{K} + \frac{k}{K})\right]\left[\prod_{k=0}^{K-1} \Gamma(\frac{1}{K}\theta + \frac{1}{K}\beta + \frac{k}{K})\right]}{\left[\prod_{k=0}^{K-1} \Gamma(\frac{1}{K}\theta + \frac{1}{K}\beta + \frac{r}{K} + \frac{k}{K})\right]\left[\prod_{k=0}^{K-1} \Gamma(\frac{1}{K}\theta + \frac{k}{K})\right]} \\
&= \frac{\left[(2\pi)^{\frac{K-1}{2}} K^{\frac{1}{2}-(\theta+r)} \Gamma(\theta + r)\right]\left[(2\pi)^{\frac{K-1}{2}} K^{\frac{1}{2}-(\theta+\beta)} \Gamma(\theta + \beta)\right]}{\left[(2\pi)^{\frac{K-1}{2}} K^{\frac{1}{2}-(\theta+\beta+r)} \Gamma(\theta + \beta + r)\right]\left[(2\pi)^{\frac{K-1}{2}} K^{\frac{1}{2}-\theta} \Gamma(\theta)\right]} \\
&= \frac{\Gamma(\theta + r)\Gamma(\beta)\Gamma(\theta + \beta)}{\Gamma(\theta + \beta + r)\Gamma(\theta)\Gamma(\beta)} \\
&= \frac{B(\theta + r, \beta)}{B(\theta, \beta)}.
\end{aligned}$$

This matches the moments of a Beta(θ, β) distribution, implying that $\left(\prod_{k=1}^{K} X^{(k)}\right)^{\frac{1}{K}} \sim$ Beta(θ, β) as required.

2. *A sufficient statistic for θ in the joint distribution of $X^{(1)}, \ldots, X^{(K)}$ is*

$$T(X^{(1)}, \ldots, X^{(K)}) = \left(\prod_{k=1}^{K} X^{(k)}\right)^{\frac{1}{K}}.$$

By the mutual independence of $X^{(k)}$, the joint density of $X^{(1)}, \ldots, X^{(K)}$ can be written as

$$\begin{aligned} f_{X^{(1)},\ldots,X^{(K)}}(x^{(1)}, \ldots, x^{(K)}) &= \prod_{k=1}^{K} f_{X^{(k)}}(x^{(k)}) \\ &\propto \prod_{k=1}^{K} \left(x^{(k)}\right)^{\frac{1}{K}\theta + \frac{k-1}{K} - 1} \left(1 - x^{(k)}\right)^{\frac{1}{K}\beta - 1} \\ &= \left[\left(\prod_{k=1}^{K} x^{(k)}\right)^{\frac{1}{K}}\right]^{\theta} \left[\prod_{k=1}^{K} \left(x^{(k)}\right)^{\frac{k-1}{K} - 1}\right] \left[\prod_{k=1}^{K} \left(1 - x^{(k)}\right)\right]^{\frac{1}{K}\beta - 1}. \end{aligned}$$

By the factorization theorem, $T(X^{(1)}, \ldots, X^{(K)}) = \left(\prod_{k=1}^{K} X^{(k)}\right)^{\frac{1}{K}}$ is a sufficient statistic for θ.

3. *To sample from G_t, i.e., the distribution of $(X^{(1)}, \ldots, X^{(K)})|T(X^{(1)}, \ldots, X^{(K)}) = t$, we first sample from $(X^{(1)}, \ldots, X^{(K-1)})|T(X^{(1)}, \ldots, X^{(K)}) = t$ and then recover $X^{(K)}$.*

We will show that the conditional density $f_{X^{(1)},\ldots,X^{(K-1)}|T(X^{(1)},\ldots,X^{(K)})=t}(x^{(1)}, \ldots, x^{(K-1)})$ is, up to a normalizing constant involving t,

$$\left[\prod_{k=1}^{K-1} \left(x^{(k)}\right)^{\frac{k-K}{K} - 1}\right] \left[\left(\prod_{k=1}^{K-1} \left(1 - x^{(k)}\right)\right)\left(1 - \frac{t^K}{\prod_{k=1}^{K-1} x^{(k)}}\right)\right]^{\frac{1}{K}\beta - 1}.$$

We derive this as follows (where any factors not involving $x^{(1)}, \ldots, x^{(K-1)}$ are omitted, and we write $\theta_k = \frac{\theta}{K} + \frac{k-1}{K}$):

$$
\begin{aligned}
&f_{X^{(1)},\ldots,X^{(K-1)}|T(X^{(1)},\ldots,X^{(K)})=t}(x^{(1)},\ldots,x^{(K-1)}) \\
&\propto f_{X^{(1)},\ldots,X^{(K-1)},T(X^{(1)},\ldots,X^{(K)})}(x^{(1)},\ldots,x^{(K-1)},t) \\
&= f_{X^{(1)},\ldots,X^{(K)}}\left(x^{(1)},\ldots,x^{(K-1)},\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)\left|\frac{\partial}{\partial t}\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right| \\
&= \left(\prod_{k=1}^{K-1} f_{X^{(k)}}\left(x^{(k)}\right)\right) f_{X^{(K)}}\left(\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)\left|\frac{Kt^{K-1}}{\prod_{k=1}^{K-1}x^{(k)}}\right| \\
&\propto \left(\prod_{k=1}^{K-1}\left(x^{(k)}\right)^{\theta_k-1}\left(1-x^{(k)}\right)^{\frac{1}{K}\beta-1}\right)\left(\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)^{\theta_K-1}\left(1-\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)^{\frac{1}{K}\beta-1}\frac{1}{\prod_{k=1}^{K-1}x^{(k)}} \\
&\propto \left[\prod_{k=1}^{K-1}\left(x^{(k)}\right)^{\theta_k-\theta_K-1}\right]\left[\left(\prod_{k=1}^{K-1}\left(1-x^{(k)}\right)\right)\left(1-\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)\right]^{\frac{1}{K}\beta-1}.
\end{aligned}
$$

It remains to note that $\theta_k - \theta_K = \frac{k-K}{K}$.

To generate $(X^{(1)},\ldots,X^{(K)})$, first sample from $(X^{(1)},\ldots,X^{(K-1)})|T(X^{(1)},\ldots,X^{(K)}) = t$ with numerical sampling methods. In this example, a Metropolis algorithm with a uniform proposal over $[t^K,1)^K$ is effective (Metropolis *et al.*, 1953). Then, compute $X^{(K)} = \frac{t^K}{\prod_{k=1}^{K-1}X^{(k)}}$.

□

C.1.3 Dirichlet distribution

The Dirichlet distribution (which subsumes the beta distribution) on the K-simplex is typically parameterized by a K-dimensional vector $\boldsymbol{\alpha}$. It can also be parameterized by the mean, defined as $\boldsymbol{\alpha}/\sum_{k=1}^K \alpha_k$, and precision, defined as $\sum_{k=1}^K \alpha_k$. Using the mean-precision parameterization, the Dirichlet distribution with known precision ϕ and unknown mean $\boldsymbol{\theta}$ can be thinned into K gamma random variables.

Example C.2 (Thinning Dirichlet$_K(\boldsymbol{\theta},\phi)$ with ϕ known)**.** *Following the steps of Algorithm 1, start with K mutually independent gamma random variables, $X^{(k)} \sim Gamma(\theta_k\phi,\nu)$ for $k = 1,\ldots,K$ where $\nu > 0$ is a tuning parameter chosen by the user. Then, note that*

$$
T(X^{(1)},\ldots,X^{(K)}) = (X^{(1)},\ldots,X^{(K)})^\top / \sum_{k=1}^K X^{(K)}
$$

is a sufficient statistic for $\boldsymbol{\theta}$ on the basis of $(X^{(1)},\ldots,X^{(K)})$. Since $T(X^{(1)},\ldots,X^{(K)}) \sim Dirichlet_K(\boldsymbol{\theta},\phi)$, we can thus thin the Dirichlet distribution into $(X^{(1)},\ldots,X^{(K)})$.

To generate $(X^{(1)},\ldots,X^{(K)})$, first sample from the conditional distribution of $X^{(1)}$ given $T(X^{(1)},\ldots,X^{(K)}) = \mathbf{t}$. This follows a Gamma$(\phi,\nu/t_1)$ distribution. Then for $k = 2,\ldots,K$, set $X^{(k)} = X^{(1)}t_k/t_1$.

Proof of Example C.2. We must prove that $(X^{(1)}, \ldots, X^{(K)})^\top / \sum_{k=1}^K X^{(K)}$ is a sufficient statistic for $\boldsymbol{\theta}$ in the joint distribution of $X^{(k)} \sim \text{Gamma}(\theta_k \phi, \nu)$ for $k = 1, \ldots, K$.

Consider the joint density of $X^{(k)}$ for $k = 1, \ldots, K$:

$$\begin{aligned}
f_{X^{(1)},\ldots,X^{(K)}}(x^{(1)}, \ldots, x^{(K)}) &= \prod_{k=1}^K f_{X^{(k)}}(x^{(k)}) \\
&\propto \left[\prod_{k=1}^K \left(x^{(k)}\right)^{\theta_k \phi - 1} \exp\left(-\nu x^{(k)}\right) \right] \\
&= \prod_{k=1}^K \left[\left(x^{(k)}\right)^{\theta_k \phi - 1} \left(\sum_{k'=1}^K x^{(k')} \right)^{(\theta_k \phi - 1) - (\theta_k \phi - 1)} \exp\left(-\nu x^{(k)}\right) \right] \\
&= \prod_{k=1}^K \left[\left(\frac{x^{(k)}}{\sum_{k'=1}^K x^{(k')}} \right)^{\theta_k \phi - 1} \left(\sum_{k'=1}^K x^{(k')} \right)^{\theta_k \phi - 1} \exp\left(-\nu x^{(k)}\right) \right] \\
&= \left(\sum_{k'=1}^K x^{(k')} \right)^{\sum_{k=1}^K (\theta_k \phi - 1)} \prod_{k=1}^K \left[\left(\frac{x^{(k)}}{\sum_{k'=1}^K x^{(k')}} \right)^{\theta_k \phi - 1} \exp\left(-\nu x^{(k)}\right) \right] \\
&= \left(\sum_{k'=1}^K x^{(k')} \right)^{\phi \sum_{k=1}^K \theta_k - K} \prod_{k=1}^K \left[\left(\frac{x^{(k)}}{\sum_{k'=1}^K x^{(k')}} \right)^{\theta_k \phi - 1} \exp\left(-\nu x^{(k)}\right) \right] \\
&= \left(\sum_{k'=1}^K x^{(k')} \right)^{\phi - K} \prod_{k=1}^K \left[\left(\frac{x^{(k)}}{\sum_{k'=1}^K x^{(k')}} \right)^{\theta_k \phi - 1} \exp\left(-\nu x^{(k)}\right) \right]
\end{aligned}$$

The above implies by the factorization theorem that $(X^{(1)}, \ldots, X^{(K)})^\top / \sum_{k=1}^K X^{(K)}$ is a sufficient statistic for $\boldsymbol{\theta}$ on the basis of $(X^{(1)}, \ldots, X^{(K)})$ as required.

□

C.1.4 Gamma distribution

In proving Example 4.2, rather than work with gamma random variables directly, we will find it convenient to work with the logarithm of gamma random variables. We start by deriving the moment generating function of this distribution.

Lemma 1. *Consider a random variable Y such that $e^Y \sim$ Gamma(θ, β). Then the moment generating function of Y exists in a neighborhood around 0 and is given by*

$$\Phi_Y(t) = \frac{\Gamma(\theta + t)}{\Gamma(\theta)\beta^t}.$$

Proof of Lemma 1. The density of Y is given by

$$\begin{aligned} f_Y(y) &= f_{\text{Gamma}(\theta,\beta)}(e^y)e^y \\ &= \frac{\beta^\theta}{\Gamma(\theta)} e^{y(\theta-1)} e^{-\beta e^y} e^y \\ &= \frac{\beta^\theta}{\Gamma(\theta)} e^{y\theta - \beta e^y}, \end{aligned}$$

where the extra factor of e^y in the first equality is the Jacobian of the transformation. For $t > -\theta$, the moment generating function for this random variable is given by

$$\begin{aligned} \Phi_Y(t) &= \mathbb{E}[e^{tY}] \\ &= \int e^{ty} \frac{\beta^\theta}{\Gamma(\theta)} e^{y\theta - \beta e^y} dy \\ &= \frac{\beta^\theta}{\Gamma(\theta)} \frac{\Gamma(\theta+t)}{\beta^{\theta+t}} \int \frac{\beta^{\theta+t}}{\Gamma(\theta+t)} e^{y(\theta+t) - \beta e^y} dy \\ &= \frac{\Gamma(\theta+t)}{\Gamma(\theta)\beta^t} \int \frac{\beta^{\theta+t}}{\Gamma(\theta+t)} e^{y(\theta+t) - \beta e^y} dy \\ &= \frac{\Gamma(\theta+t)}{\Gamma(\theta)\beta^t}. \end{aligned}$$

The assumption that $t > -\theta$ ensures that $\theta + t > 0$ so that the integrand in the second-to-last line is the density of the logarithm of a Gamma$(\theta + t, \beta)$ random variable. Since $\theta > 0$, we have established that the moment generating function exists in a neighborhood around 0. □

Proof of Example 4.2. We must prove the following three claims:

1. *If for* $k = 1, \ldots, K$, $X^{(k)} \sim$ *Gamma* $\left(\frac{1}{K}\theta + \frac{k-1}{K}, \frac{1}{K}\beta\right)$ *and* $X^{(k)}$ *are mutually independent, then* $\left(\prod_{k=1}^K X^{(k)}\right)^{\frac{1}{K}} \sim$ *Gamma*(θ, β).

   Defining $Y^{(k)} = \log X^{(k)}$ and $\bar{Y}_K = \frac{1}{K}\sum_{k=1}^K Y^{(k)}$, observe that

   $$\left(\prod_{k=1}^K X^{(k)}\right)^{\frac{1}{K}} = e^{\bar{Y}_K}.$$

   Thus, our goal is to prove that $e^{\bar{Y}_K} \sim \text{Gamma}(\theta, \beta)$. Since the moment generating function completely characterizes a distribution, it is sufficient to show that $\Phi_{\bar{Y}_K}(t)$ matches the expression in Lemma 1. Applying Lemma 1 to $e^{Y^{(k)}} \sim \text{Gamma}(\theta_k, \beta_k)$, where $\theta_k = \theta/K + (k-1)/K$ and $\beta_k = \beta/K$, implies that

   $$\Phi_{Y^{(k)}} = \frac{\Gamma(\theta_k + t)}{\Gamma(\theta_k)\beta_k^t}.$$

By independence of $Y^{(1)}, \ldots, Y^{(K)}$ and standard properties of the moment generating function,

$$\begin{aligned}\Phi_{\bar{Y}_K}(t) &= \prod_{k=1}^{K} \Phi_{Y_K/K}(t) \\ &= \prod_{k=1}^{K} \Phi_{Y_K}(t/K) \\ &= \prod_{k=1}^{K} \frac{\Gamma(\theta_k + t/K)}{\Gamma(\theta_k)\beta_k^{t/K}}.\end{aligned}$$

Recalling the form of θ_k and β_k and applying the Gauss multiplication theorem (Abramowitz *et al.*, 1972, page 256) to both the numerator and denominator gives

$$\Phi_{\bar{Y}_K}(t) = \frac{K^{-(\theta+t)}\Gamma(\theta+t)}{K^{-\theta}\Gamma(\theta)} \frac{1}{(\beta/K)^t} = \frac{\Gamma(\theta+t)}{\Gamma(\theta)\beta^t}.$$

This completes the proof.

2. *A sufficient statistic for θ in the joint distribution of $X^{(1)}, \ldots, X^{(K)}$ is $T(X^{(1)}, \ldots, X^{(K)}) = \left(\prod_{k=1}^{K} X^{(k)}\right)^{\frac{1}{K}}$.*

   By the mutual independence of $X^{(k)}$, the joint density of $X^{(1)}, \ldots, X^{(K)}$ can be written as,

$$\begin{aligned}f_{X^{(1)},\ldots,X^{(K)}}(x^{(1)}, \ldots, x^{(K)}) &= \prod_{k=1}^{K} f_{X^{(k)}}(x^{(k)}) \\ &\propto \prod_{k=1}^{K} \left(x^{(k)}\right)^{\frac{1}{K}\theta + \frac{k-1}{K} - 1} \exp\left(-\frac{\beta}{K} x^{(k)}\right) \\ &= \left[\left(\prod_{k=1}^{K} x^{(k)}\right)^{\frac{1}{K}}\right]^{\theta} \left[\prod_{k=1}^{K} \left(x^{(k)}\right)^{\frac{k-1}{K} - 1}\right] \exp\left(-\frac{\beta}{K} \sum_{k=1}^{K} x^{(k)}\right)\end{aligned}$$

   By the factorization theorem, $T(X^{(1)}, \ldots, X^{(K)}) = \left(\prod_{k=1}^{K} X^{(k)}\right)^{\frac{1}{K}}$ is a sufficient statistic for θ as required.

3. *To sample from G_t, the conditional distribution $(X^{(1)}, \ldots, X^{(K)})|T(X^{(1)}, \ldots, X^{(K)}) = t$, we first sample from $(X^{(1)}, \ldots, X^{(K-1)})|T(X^{(1)}, \ldots, X^{(K)}) = t$ and then recover $X^{(K)}$.*

The conditional density $f_{X^{(1)},\ldots,X^{(K-1)}|T(X^{(1)},\ldots,X^{(K)})=t}(x^{(1)},\ldots,x^{(K-1)})$, up to a normalizing constant depending on t, is

$$\left[\prod_{k=1}^{K-1}\left(x^{(k)}\right)^{\frac{k-K}{K}-1}\right]\exp\left(-\frac{\beta}{K}\left(\sum_{k=1}^{K-1}x^{(k)}+\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)\right).$$

The derivation is as follows (where any factors not involving $x^{(1)},\ldots,x^{(K-1)}$ are omitted and we write $\theta_k=\frac{\theta}{K}+\frac{k-1}{K}$):

$$\begin{aligned}
&f_{X^{(1)},\ldots,X^{(K-1)}|T(X^{(1)},\ldots,X^{(K)})=t}(x^{(1)},\ldots,x^{(K-1)})\\
&\propto f_{X^{(1)},\ldots,X^{(K-1)},T(X^{(1)},\ldots,X^{(K)})}(x^{(1)},\ldots,x^{(K-1)},t)\\
&=f_{X^{(1)},\ldots,X^{(K)}}\left(x^{(1)},\ldots,x^{(K-1)},\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)\left|\frac{\partial}{\partial t}\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right|\\
&=\left(\prod_{k=1}^{K-1}f_{X^{(k)}}\left(x^{(k)}\right)\right)f_{X^{(K)}}\left(\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)\left|\frac{Kt^{K-1}}{\prod_{k=1}^{K-1}x^{(k)}}\right|\\
&\propto\left(\prod_{k=1}^{K-1}\left(x^{(k)}\right)^{\theta_k-1}\exp\left(-\frac{\beta}{K}x^{(k)}\right)\right)\left(\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)^{\theta_K-1}\exp\left(-\frac{\beta}{K}\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)\frac{1}{\prod_{k=1}^{K-1}x^{(k)}}\\
&\propto\left(\prod_{k=1}^{K-1}\left(x^{(k)}\right)^{\frac{k-K}{K}-1}\right)\exp\left(-\frac{\beta}{K}\left(\sum_{k=1}^{K-1}x^{(k)}+\frac{t^K}{\prod_{k=1}^{K-1}x^{(k)}}\right)\right),
\end{aligned}$$

where in the last step we used that $\theta_k-\theta_K=\frac{k-K}{K}$.

To generate $(X^{(1)},\ldots,X^{(K)})$, first sample from $(X^{(1)},\ldots,X^{(K-1)})|T(X^{(1)},\ldots,X^{(K)})=t$ with numerical sampling methods. In this example, MCMC methods work well though the choice of proposal distribution should consider K and β. Then, compute $X^{(K)}=\frac{t^K}{\prod_{k=1}^{K-1}X^{(k)}}$.

□

C.2 Families with support controlled by an unknown parameter

C.2.1 Scaled beta distribution

We consider the family of distributions obtained by scaling a Beta$(\alpha,1)$ distribution (with α fixed) by an unknown scale parameter $\theta>0$. In the special case that $\alpha=1$, this corresponds to the Unif$(0,\theta)$ family presented in Example 5.1.

Example C.3 (Thinning $\theta\cdot\text{Beta}(\alpha,1)$ with α known)**.** *We start with* $X^{(k)}\overset{iid}{\sim}\theta\cdot Beta\left(\frac{\alpha}{K},1\right)$ *for* $k=1,\ldots,K$*, and note that* $T(X^{(1)},\ldots,X^{(K)})=\max(X^{(1)},\ldots,X^{(K)})$ *is sufficient for* θ*.*

Furthermore, $\max(X^{(1)}, \ldots, X^{(K)}) \sim \theta \cdot Beta(\alpha, 1)$*. Thus, we define* G_t *to be the conditional distribution of* $(X^{(1)}, \ldots, X^{(K)})$ *given* $\max(X^{(1)}, \ldots, X^{(K)}) = t$*. Then, by Theorem 1, we can thin* $X \sim \theta \cdot Beta(\alpha, 1)$ *by sampling from* G_X*.*

To sample from this conditional distribution, we first draw $\mathbf{C} \sim Categorical_K(1/K, \ldots, 1/K)$*. Then,* $X^{(k)} = C_k X + (1 - C_k) Z_k$ *where* $Z_k \overset{iid}{\sim} X \cdot Beta\left(\frac{\alpha}{K}, 1\right)$*.*

Proof of Example C.3. We must prove the following three claims:

1. *If for* $k = 1, \ldots, K$*,* $X^{(k)} \overset{iid}{\sim} \theta \cdot Beta\left(\frac{\alpha}{K}, 1\right)$*, then* $\max(X^{(1)}, \ldots, X^{(K)}) \sim \theta \cdot Beta(\alpha, 1)$*.*

   First, note that $\frac{1}{\theta} X^{(k)} \overset{iid}{\sim} \text{Beta}\left(\frac{\alpha}{K}, 1\right)$. The distribution of $\max(X^{(1)}, \ldots, X^{(K)})$ can be derived using the CDF method as follows:

$$\begin{aligned} P(\max(X^{(1)}, \ldots, X^{(K)}) \leq z) &= P(X^{(1)} \leq z, \ldots, X^{(K)} \leq z) \\ &= \prod_{k=1}^{K} P(X^{(k)} \leq z) \\ &= \prod_{k=1}^{K} P\left(\frac{1}{\theta} X^{(k)} \leq \frac{z}{\theta}\right) \\ &= \prod_{k=1}^{K} \left(\frac{z}{\theta}\right)^{\frac{\alpha}{K}} \\ &= \left(\frac{z}{\theta}\right)^{\alpha}, \end{aligned}$$

   where we have used that $P(\text{Beta}(\alpha, 1) \leq x) = x^\alpha$ for $x \in (0, 1)$. The above implies that $\max(X^{(1)}, \ldots, X^{(K)}) \sim \theta \cdot \text{Beta}(\alpha, 1)$ as required.

2. *A sufficient statistic for* θ *based on* $X^{(1)}, \ldots, X^{(K)}$ *is* $\max(X^{(1)}, \ldots, X^{(K)})$*.*

   Using the independence of $X^{(k)}$, the joint distribution can be written as

$$\begin{aligned} & f_{X^{(1)}, \ldots, X^{(K)}}(x^{(1)}, \ldots, x^{(K)}) \\ &= \prod_{k=1}^{K} f_{X^{(k)}}(x^{(k)}) \\ &\propto \prod_{k=1}^{K} \left(x^{(k)}\right)^{\frac{\alpha}{K} - 1} I\{0 < x^{(k)} < \theta\} \\ &= \left(\prod_{k=1}^{K} x^{(k)}\right)^{\frac{\alpha}{K} - 1} I\{\min(x^{(1)}, \ldots, x^{(K)}) > 0\} I\{\max(x^{(1)}, \ldots, x^{(K)}) < \theta\}. \end{aligned}$$

   By the factorization theorem, we conclude that $\max(X^{(1)}, \ldots, X^{(K)})$ is a sufficient statistic for θ as required.

3. *We can sample from the conditional distribution* $(X^{(1)}, \ldots, X^{(K)})|\max(X^{(1)}, \ldots, X^{(K)}) = t$ *by taking* $X^{(k)} = C_k t + (1 - C_k)Z_k$ *where* $\mathbf{C} \sim$ *Categorical*$_K(1/K, \ldots, 1/K)$ *and* $Z_k \sim t \cdot$ *Beta*$(\frac{\alpha}{K}, 1)$.

   Without loss of generality, consider $X^{(k)}$. Given that $X^{(1)}, \ldots, X^{(K)}$ are identically distributed, $P(X^{(k)} = t) = \frac{1}{K}$. Hence, in the first stage, we can draw one sample, $\mathbf{C} \sim \text{Categorical}_K(1/K, \ldots, 1/K)$ to determine if $X^{(k)}$ is the maximum. If $X^{(k)}$ is not the maximum then we know that $X^{(k)} \leq t$. We can compute the density of $Z_k \stackrel{D}{=} (X^{(k)}|X^{(k)} \leq t)$ as follows,

$$f_{X^{(k)}|X^{(k)} \leq t}(x^{(k)}) = \frac{f_{X^{(k)}}(x^{(k)})}{P(X^{(k)} \leq t)} = \frac{\frac{1}{\theta^{\frac{\alpha}{K}} B\left(\frac{\alpha}{K}, 1\right)} \left(x^{(k)}\right)^{\frac{\alpha}{K} - 1}}{\left(\frac{t}{\theta}\right)^{\frac{\alpha}{K}}} = \frac{1}{t^{\frac{\alpha}{K}} B\left(\frac{\alpha}{K}, 1\right)} \left(x^{(k)}\right)^{\frac{\alpha}{K} - 1}.$$

   The above implies that $Z_k \sim t \cdot \text{Beta}\left(\frac{\alpha}{K}, 1\right)$ as required.

The result then follows from Theorem 1. □

C.2.2 Shifted exponential distribution

We consider $X \sim \text{SExp}(\theta, \lambda)$, which is the location family generated by shifting an exponential random variable by an amount θ. It has density

$$p_{\theta,\lambda}(x) = \lambda e^{-\lambda(x-\theta)} 1\{x \geq \theta\}.$$

Example C.4 (Thinning a $\text{SExp}(\theta, \lambda)$ random variable with known λ)**.** *We begin with* $X^{(k)} \stackrel{iid}{\sim} \text{SExp}(\theta, \lambda/K)$ *for* $k = 1, \ldots, K$*, and note that* $T(X^{(1)}, \ldots, X^{(K)}) = \min(X^{(1)}, \ldots, X^{(K)})$ *is sufficient for* θ*. Furthermore,* $\min(X^{(1)}, \ldots, X^{(K)}) \sim \text{SExp}(\theta, \lambda)$*. We define* G_t *to be the conditional distribution of* $(X^{(1)}, \ldots, X^{(K)})$ *given* $\min(X^{(1)}, \ldots, X^{(K)}) = t$*. Then, by Theorem 1, we can thin* $X \sim \text{SExp}(\theta, \lambda)$ *by sampling from* G_X*.*

To sample from G_X*, we first draw* $\mathbf{C} \sim$ *Categorical*$_K(1/K, \ldots, 1/K)$*. We then take* $X^{(k)} = X + (1 - C_k)Z_k$*, where* $Z_k \stackrel{iid}{\sim} \text{Exp}(\lambda/K)$*.*

Proof of Example C.4. We must prove the following three claims:

1. *If for* $k = 1, \ldots, K$*,* $X^{(k)} \stackrel{iid}{\sim}$ *SExp*$(\theta, \lambda/K)$*, then* $\min(X^{(1)}, \ldots, X^{(K)}) \sim$ *SExp*(θ, λ)*.*

   First, note that $X^{(k)} - \theta \stackrel{iid}{\sim} \text{Exp}(\lambda/K)$. The distribution of $\min(X^{(1)}, \ldots, X^{(K)})$ can

be derived using the CDF method as follows,

$$\begin{aligned}
P(\min(X^{(1)},\ldots,X^{(K)}) \geq z) &= P(X^{(1)} \geq z,\ldots,X^{(K)} \geq z) \\
&= \prod_{k=1}^{K} P(X^{(k)} \geq z) \\
&= \prod_{k=1}^{K} P\left(X^{(k)} - \theta \geq z - \theta\right) \\
&= \prod_{k=1}^{K} \exp\left(-\frac{\lambda}{K}(z-\theta)\right) \\
&= \exp\left(-\lambda(z-\theta)\right).
\end{aligned}$$

The above implies that $\min(X^{(1)},\ldots,X^{(K)}) \sim \text{SExp}(\theta,\lambda)$ as required.

2. *A sufficient statistic for θ based on $X^{(1)},\ldots,X^{(K)}$ is* $\min(X^{(1)},\ldots,X^{(K)})$.
Using the independence of $X^{(k)}$, the joint distribution can be written as

$$\begin{aligned}
&f_{X^{(1)},\ldots,X^{(K)}}(x^{(1)},\ldots,x^{(K)}) \\
&= \prod_{k=1}^{K} f_{X^{(k)}}(x^{(k)}) \\
&\propto \prod_{k=1}^{K} \exp\left(-\frac{\lambda}{K}(x^{(k)}-\theta)\right) I\{x^{(k)} > \theta\} \\
&\propto \exp\left(-\frac{\lambda}{K}\sum_{k=1}^{K} x^{(k)}\right) I\{\min(x^{(1)},\ldots,x^{(K)}) > \theta\}.
\end{aligned}$$

Given that the joint distribution can be written such that θ only interacts with the data through the $I\{\min(x^{(1)},\ldots,x^{(K)}) > \theta\}$ term, we conclude that $\min(X^{(1)},\ldots,X^{(K)})$ is a sufficient statistic for θ as required.

3. *We can sample from the conditional distribution $\left(X^{(1)},\ldots,X^{(K)}\right) \mid \min(X^{(1)},\ldots,X^{(K)}) = t$ by taking $X^{(k)} = t + (1-C_k)Z_k$ where $\mathbf{C} \sim$ Categorical$_K(1/K,\ldots,1/K)$ and $Z_k \sim$ Exp(λ/K).*

Without loss of generality, consider $X^{(k)}$. Given that $X^{(1)},\ldots,X^{(K)}$ are identically distributed, $P(X^{(k)} = X) = \frac{1}{K}$. Hence, in the first stage, we can draw one sample, $\mathbf{C} \sim \text{Categorical}_K(1/K,\ldots,1/K)$ to determine if $X^{(k)}$ is the minimum. Otherwise, we require that $X^{(k)} \geq t$. We know that the density of $Z_k \overset{D}{=} (X^{(k)} | X^{(k)} \geq t) \overset{D}{=} X^{(k)}$ by the memoryless property of the exponential distribution. Thus, $Z_k \sim \text{Exp}\,(\lambda/K)$ as required.

The result then follows from Theorem 1.

□

D Additional example of indirect thinning

We consider an example of indirect thinning in which $S(\cdot)$ is neither invertible, nor scalar-valued. Specifically, let $\mathbf{X} = (X_1, \ldots, X_n)$ represent a sample of n independent and identically distributed normal observations with unknown mean θ_1 and variance θ_2. Using ideas from Theorem 2 and Proposition 1, we thin $\mathbf{X}$ through the sample mean and sample variance.

Example D.1 (Indirect thinning of $N_n(\theta_1 \mathbf{1}_n, \theta_2 \mathbf{I}_n)$ through the sample mean and sample variance)**.** *Suppose $\mathbf{X} \sim N_n(\theta_1 \mathbf{1}_n, \theta_2 \mathbf{I}_n)$. Then $S(\mathbf{X})$ is sufficient for $\boldsymbol{\theta} = (\theta_1, \theta_2)$, where*

$$S(\mathbf{x}) = \left(\frac{1}{n} \sum_{i=1}^{n} x_i, \frac{1}{n-1} \sum_{i=1}^{n} \left(x_i - \frac{1}{n} \sum_{i'=1}^{n} x_{i'} \right)^2 \right).$$

We will indirectly thin $\mathbf{X}$ through $S(\cdot)$ into $K = n$ univariate normals. To do so, we start with $X^{(k)} \overset{iid}{\sim} N(\theta_1, \theta_2)$ for $k = 1, \ldots, K$. A sufficient statistic for $\boldsymbol{\theta}$ based on $(X^{(1)}, \ldots, X^{(K)})$ is $T(X^{(1)}, \ldots, X^{(K)})$, where $T(x^{(1)}, \ldots, x^{(K)}) = S((x^{(1)}, \ldots, x^{(K)})^\top)$, i.e., we concatenate the K entries into a vector and apply $S(\cdot)$. Furthermore, $T(X^{(1)}, \ldots, X^{(K)})$ has the same distribution as $S(\mathbf{X})$, since $(X^{(1)}, \ldots, X^{(K)})^\top$ and $\mathbf{X}$ have the same distribution. This establishes that we can indirectly thin $\mathbf{X}$ through $S(\cdot)$ by $T(\cdot)$.

By Theorem 1, we define $G_{\mathbf{t}}$ to be the conditional distribution of $(X^{(1)}, \ldots, X^{(K)})$ given $T(X^{(1)}, \ldots, X^{(K)}) = \mathbf{t}$, i.e. it is the distribution of a $N_K(\theta_1 \mathbf{1}_K, \theta_2 \mathbf{I}_K)$ random vector conditional on its sample mean and sample variance equalling $\mathbf{t} = (t_1, t_2)$. This conditional distribution is uniform over the set of points in $\mathbb{R}^K$ with sample mean t_1 and sample variance t_2. To see this, note that $G_{\mathbf{t}}$ cannot depend on $\boldsymbol{\theta}$ (by sufficiency), so we can take $\boldsymbol{\theta} = (t_1, t_2)$ and equivalently describe $G_{\mathbf{t}}$ as the distribution of a $N_K(t_1 \mathbf{1}_K, t_2 \mathbf{I}_K)$ random vector conditional on its sample mean and sample variance equalling $\mathbf{t} = (t_1, t_2)$. Such a distribution has constant density on a sphere centered at $t_1 \mathbf{1}_K$. Thus, the conditional distribution $G_{\mathbf{t}}$ is uniform over the set of points in $\mathbb{R}^K$ with sample mean t_1 and sample variance t_2. Finally, we obtain $(X^{(1)}, \ldots, X^{(K)})$ by sampling from $G_{\mathbf{X}}$.

In effect, Example D.1 shows that given a realization of $\mathbf{X} \sim N_n(\theta_1 \mathbf{1}_n, \theta_2 \mathbf{I}_n)$, we can generate a new random vector $(X^{(1)}, \ldots, X^{(n)})^\top$ with the identical distribution, and with the same sample mean and sample variance, without knowledge of the true mean or true variance. This might have applications in cases where the true values of the observations cannot be shared. Furthermore, the ideas in Example D.1 can be applied in settings where only the sufficient statistics of a realization of $\mathbf{X} \sim N_n(\theta_1 \mathbf{1}_n, \theta_2 \mathbf{I}_n)$ are available, and we wish to generate a "plausible" sample that could have led to those sufficient statistics.

E Numerical experiments

In this section, we illustrate some of the examples from Sections 3, 4, and 5 through numerical simulations. Specifically, we thin a Gamma(α, θ) distribution using the three different

approaches described in Examples 3.2, 3.3, and C.1, a Beta(θ, β) into two non-identical beta random variables as described in Example 4.1, and a Unif$(0, \theta)$ into scaled betas as described in Example 5.1. We take $K = 2$ throughout for ease of presentation.

In Figure S1, each row corresponds to one of the examples mentioned above. In the left-hand column, we display the empirical density of $B = 100,000$ realizations, x_b (for $b = 1, \ldots, B$), of the $X \sim P_\theta$ that we wish to thin, overlaid with the true density of P_θ. In the center-left and center-right columns, we display B realizations of $X^{(1)}$ and $X^{(2)}$ respectively, where each realization $(x_b^{(1)}, x_b^{(2)})$ is obtained by sampling from G_{x_b}, the conditional distribution of $(X^{(1)}, X^{(2)})$ given $T(X^{(1)}, X^{(2)}) = x_b$. (This sampling is done without knowledge of θ.) We overlay the densities of the marginals $Q_\theta^{(1)}$ and $Q_\theta^{(2)}$. The right-hand column displays the empirical joint distribution of $(Q_\theta^{(1)}(X^{(1)}), Q_\theta^{(2)}(X^{(2)}))$.

In each case, our empirical findings corroborate our theoretical results: We see that the empirical distribution of $X \sim P_\theta$ agrees with its theoretical density (left-hand column); that the empirical distributions of $X^{(1)}$ and $X^{(2)}$ sampled from G_X coincide with $Q_\theta^{(1)}$ and $Q_\theta^{(2)}$ (even though the empirical distributions were obtained without knowledge of θ; center-left and center-right columns); and that the joint distribution of $Q_\theta^{(1)}(X^{(1)})$ and $Q_\theta^{(2)}(X^{(2)})$ resembles the independence copula (right-hand column).

F Changepoint detection simulations

First, we generate data with a common variance, specifically $X_1, \ldots, X_{2000} \overset{\text{iid}}{\sim} N(0, 1)$, and apply the three approaches to detecting and testing for a change in variance that were described in Section 7. We repeat this process 1000 times, and display the type 1 error rate in Figure 2. The naive approach does not control the type 1 error rate, while the order-preserved sample splitting and generalized data thinning approaches do.

Figure S2 displays the estimated changepoints, as well as those for which we rejected the null hypothesis of no change in variance at the Bonferroni corrected threshold, for a single realization of the simulated data. The naive approach results in a number of false positives, while the order-preserved sample splitting and generalized data thinning approaches do not.

Next, we generated data with two true changepoints: for $i = 1, \ldots, 500$, $X_i \overset{\text{iid}}{\sim} N(0, 4)$; for $i = 501, \ldots, 1500$, $X_i \overset{\text{iid}}{\sim} N(0, 25)$; and for $i = 1501, \ldots, 2000$, $X_i \overset{\text{iid}}{\sim} N(0, 1)$. We again apply the three approaches to detecting and testing for a change in variance that were described in Section 7, and display the results in Figure S3. In this setting, all three approaches reject the null hypothesis of no change in variance at two timepoints, which are located exactly at the two true changepoints.

G Relaxing the independence assumption

We now consider how the generalized data thinning recipe changes if we relax the independence requirement for $X^{(1)}$ and $X^{(2)}$.

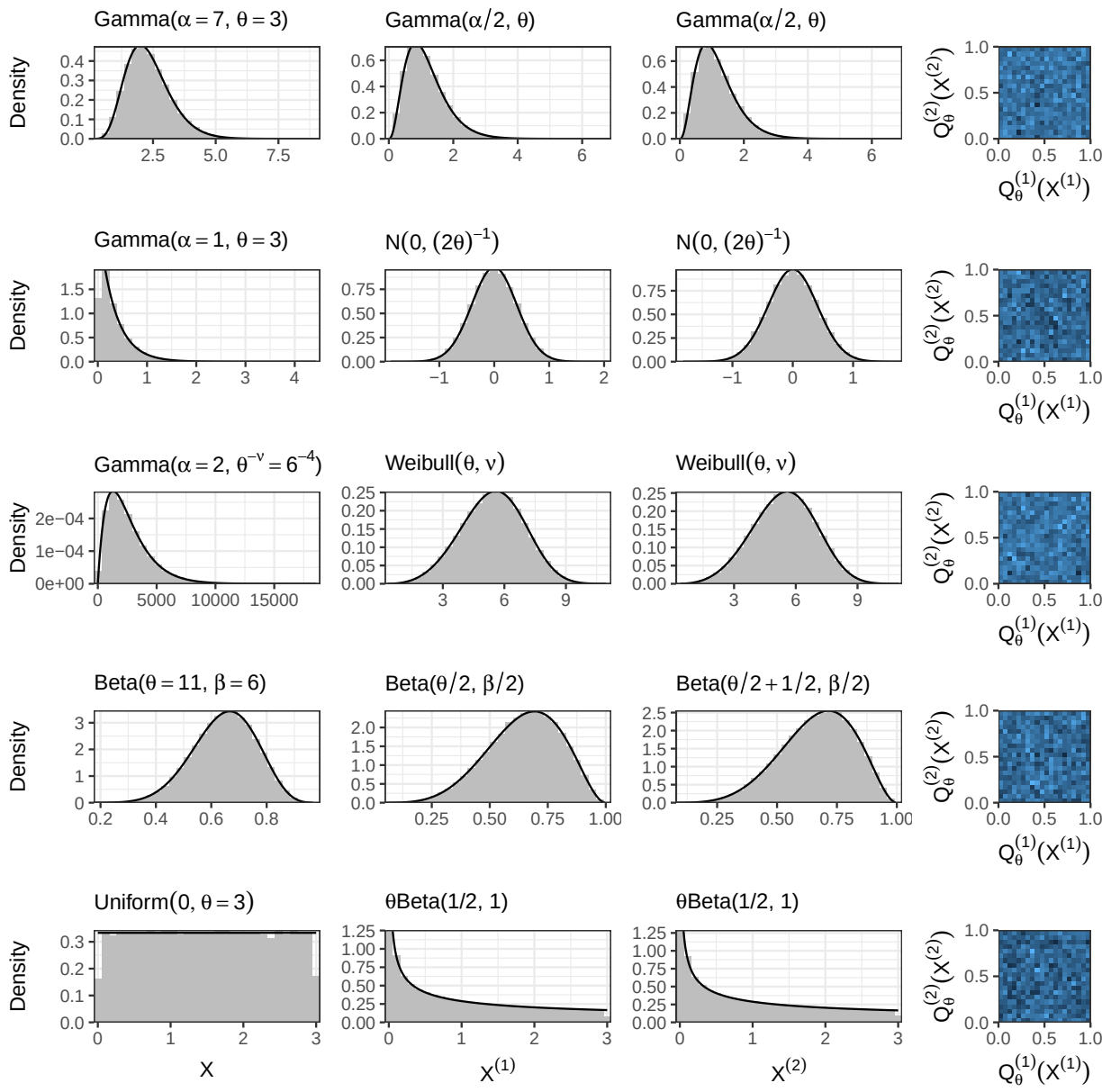


Figure S1: Numerical examples of data thinning. The left-hand column displays a sample from P_θ, which we wish to thin. The center-left and center-right columns display the empirical distributions of $X^{(1)}$ and $X^{(2)}$ that result from thinning, overlaid with the theoretical distributions $Q_\theta^{(1)}$ and $Q_\theta^{(2)}$. The right-hand column displays the empirical joint distribution of $(Q_\theta^{(1)}(X^{(1)}), Q_\theta^{(2)}(X^{(2)}))$, providing visual evidence that they are independent. With a slight abuse of notation, $Q_\theta^{(1)}(\cdot)$ and $Q_\theta^{(2)}(\cdot)$ represent the CDFs of their respective distributions.

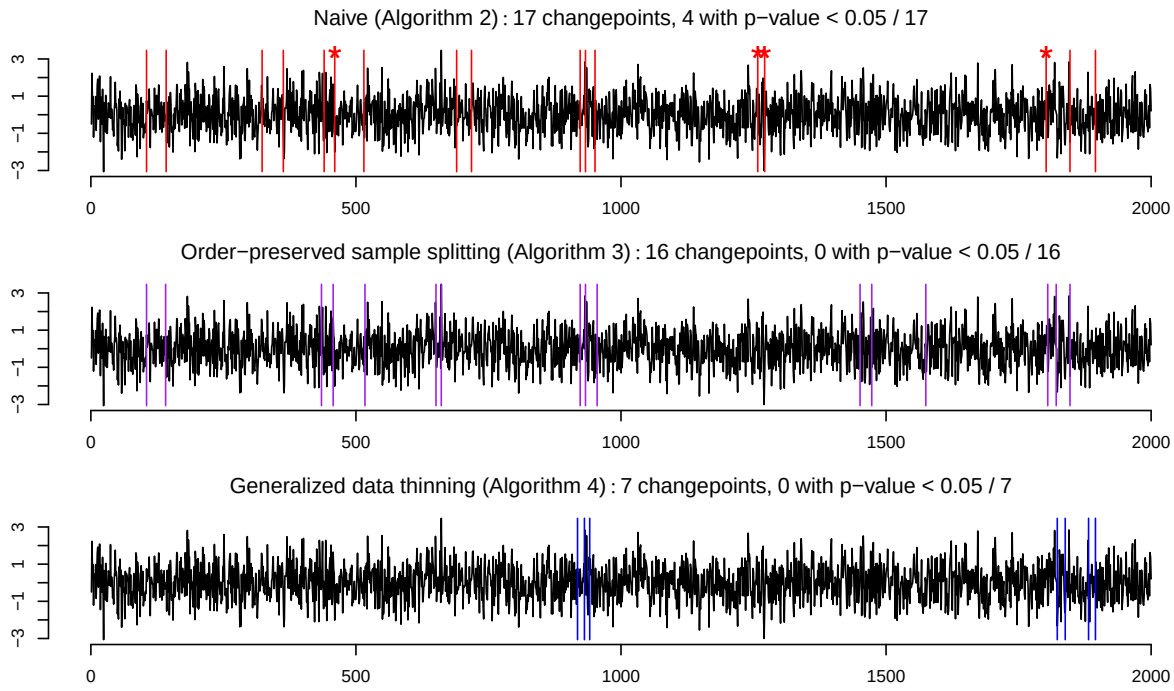


Figure S2: Results for the simulation study with no changepoints. *Top row:* Simulated data with constant variance, as well as results of the naive approach: Red lines indicate changepoints estimated using all of the data, and red asterisks indicate the estimated changepoints for which the p-value computed using all of the data was below 0.05 divided by the number of detected changepoints. The method (falsely) rejects the null hypothesis for 4 out of 17 estimated changepoints. *Middle row:* Same as the first row, but for the order-preserved sample splitting approach: Purple lines indicate changepoints estimated using the odd observations, and purple asterisks indicate those with p-values below 0.05 divided by the number of detected changepoints, using only the even observations for testing. This method (correctly) rejects none of the 16 estimated changepoints. *Bottom row:* Same as the first row, but for the generalized data thinning approach: Blue lines indicate changepoints estimated using the training set, and blue asterisks indicate those with test set p-values below 0.05 divided by the number of detected changepoints. This method (correctly) rejects none of the 7 estimated changepoints.

Algorithm 5 (Finding distributions that can be decomposed into non-independent components)**.**

1. *Choose a family of distributions* $\mathcal{Q} = \{Q_\theta : \theta \in \Omega\}$ *over* $(X^{(1)}, X^{(2)})$*, where* $X^{(1)}$ *and* $X^{(2)}$ *are not necessarily independent.*

2. *Let* $(X^{(1)}, X^{(2)}) \sim Q_\theta$*, and let* $T(X^{(1)}, X^{(2)})$ *denote a sufficient statistic for* θ*.*

3. *Let* P_θ *denote the distribution of* $T(X^{(1)}, X^{(2)})$*.*

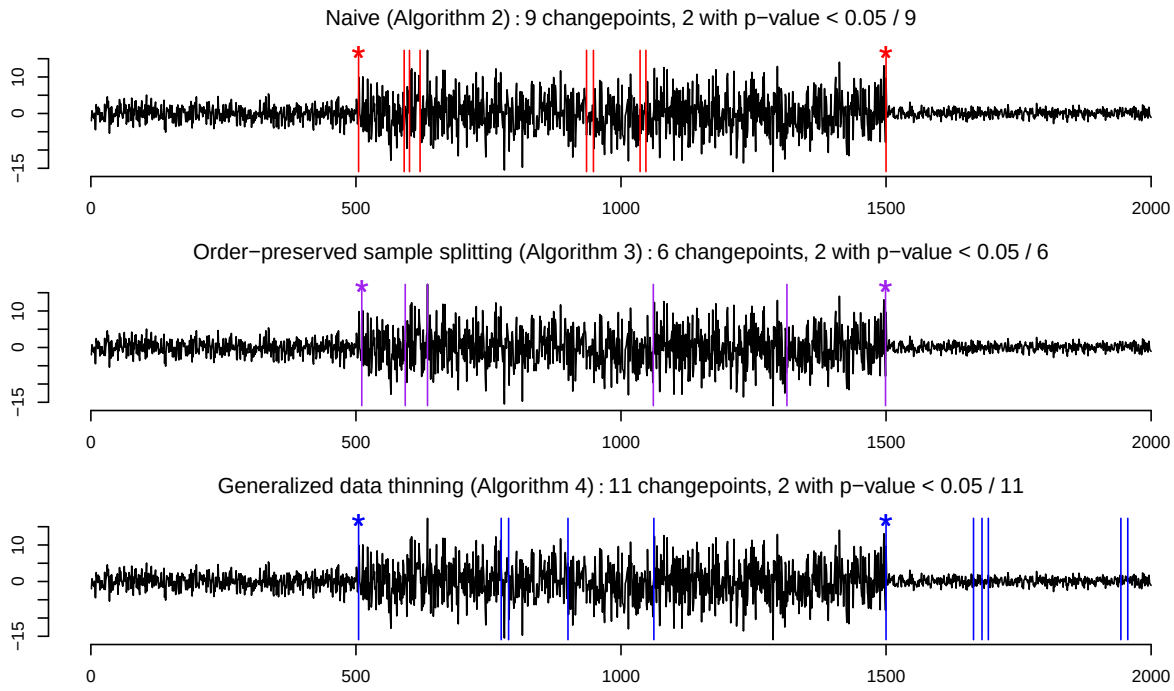


Figure S3: Results for the simulation study with changes in variance at timepoints 501 and 1501. *Top row:* Simulated data with two changepoints, as well as results of the naive approach: Red lines indicate changepoints estimated using all of the data, and red asterisks indicate the estimated changepoints for which the p-value computed using all of the data was below 0.05 divided by the number of detected changepoints. This approach leads to rejections only at the two true changepoints. *Middle row:* Same as the first row, but for the order-preserved sample splitting approach: Purple lines indicate changepoints estimated using the odd observations, and purple asterisks indicate those with p-values below 0.05 divided by the number of detected changepoints, using only the even observations for testing. This approach also leads to rejections only at the two true changepoints. *Bottom row:* Same as the first row, but for the generalized data thinning approach: Blue lines indicate changepoints estimated using the training set, and blue asterisks indicate those with test set p-values below 0.05 divided by the number of detected changepoints. This approach also leads to rejections only at the two true changepoints.

Then, given $X \sim P_\theta$, we can generate $(X^{(1)}, X^{(2)})$ by sampling from G_X, where G_t is defined as the conditional distribution

$$(X^{(1)}, X^{(2)})|T(X^{(1)}, X^{(2)}) = t.$$

By sufficiency, the sampling mechanism G_t can be performed without knowledge of θ. The key point here is that the main ideas in this paper apply even if $X^{(1)}$ and $X^{(2)}$ are dependent; however, we focused on independence in this paper to facilitate downstream

application of the decompositions that we obtain.